Worksheet 6

Statistical indices – DETAILED SOLUTIONS

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Detailed solution – Exercise 1

$$I_{2024/2023} = 115, \quad I_{2025/2024} = 110, \quad I_{2025/2023} = 126.5.$$

Circularity:

$$115 \times 110/100 = 126.5.$$

Reverse index:

$$I_{2023/2025} = 80/101.2 \times 100 = 79.0514,$$

so $126.5 \times 79.0514 \approx 10000$. An index above 100 means an increase relative to the reference period; below 100 means a decrease. **Detailed solution – Exercise 2**

The four aggregates are

$$\sum p_0q_0 = 5583, \quad \sum p_0q_1 = 6394, \quad \sum p_1q_0 = 6706, \quad \sum p_1q_1 = 7688.$$

$$L^p = 120.11, \quad P^p = 120.24, \quad F^p = \boxed{120.18},$$
$$L^q = 114.53, \quad P^q = 114.64, \quad F^q = \boxed{114.58},$$
$$I_v = \boxed{137.70}.$$

Prices rise by about 20.18%, quantities by 14.58%, and total value by 37.70%. **Detailed solution – Exercise 3**

$$\sum p_0q_0 = 2640, \quad \sum p_1q_0 = 3160, \quad \sum p_0q_1 = 2520, \quad \sum p_1q_1 = 2800.$$
$$L^p = 119.70, \quad P^p = 111.11,$$
$$F^p = \sqrt{119.70 \times 111.11} = \boxed{115.33 \text{ approximately}}.$$